



AQ5.5: Activity Questions 5 - Not Graded



This assignment will not be graded and is only for practice.

Level 1

1 point

Consider a function $T: V \rightarrow W$, where V and W are the vector spaces and following statements:

Statement 1: $T(v_1 + v_2) = T(v_1) + T(v_2)$, where $v_1, v_2 \in V$.

Statement 2: $T(cv_1) = cT(v_1)$, where $v_1 \in V$ and $c \in R$.

If T is a linear transformation then T satisfies

- ☐ statement 1 only.
- ☐ statement 2 only.
- ☐ both the statements 1 and 2.
- ☐ neither statement 1 nor statement 2.

No, the answer is incorrect.

Score: 0

Accepted Answers:

both the statements 1 and 2.

1 point

Which of the following is/are a linear transformation.

☐

$T: R^2 \rightarrow R^3, T(x, y) = (x, x + y, y + 1)$

☐

$T: R^2 \rightarrow R^2, T(x, y) = (xy, x + y)$

☐

$T: R^2 \rightarrow R^3, T(x, y) = (x + y, y, 2x + y + 1)$

☐

$T: R^2 \rightarrow R^3, T(x, y) = (x + y, y, 2x + y)$

☐

$T: R^2 \rightarrow R^2, T(x, y) = (\frac{x}{2}, \frac{y}{2})$

☐

$T: R^2 \rightarrow R^2, T(x, y) = (\frac{x}{2}, \frac{xy}{2})$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$T: R^2 \rightarrow R^3, T(x, y) = (x + y, y, 2x + y)$

$T: R^2 \rightarrow R^2, T(x, y) = (\frac{x}{2}, \frac{y}{2})$

1 point

Let $T: R^3 \rightarrow R^4$ be a linear transformation. Then which of the following options is /are true?

☐

$T(v_1 + v_2) = T(v_1) + T(v_2)$, where $v_1, v_2 \in R^3$.

☐

$T(0, 0, 0) = (0, 0, 0, 0)$

☐


$$T(0, 0, 0) = (1, 1, 1)T(0, 0, 0) = (1, 1, 1)$$

☐

$$T(3(1, 2, 0)) = 3T(1, 2, 0)T(3(1, 2, 0)) = 3T(1, 2, 0)$$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$T(v_1 + v_2) = T(v_1) + T(v_2)T(v_1 + v_2) = T(v_1) + T(v_2), \text{ where } v_1, v_2 \in R^3 v_1, v_2 \in R^3.$$

$$T(0, 0, 0) = (0, 0, 0)T(0, 0, 0) = (0, 0, 0)$$

$$T(3(1, 2, 0)) = 3T(1, 2, 0)T(3(1, 2, 0)) = 3T(1, 2, 0)$$

1 point

Let $T: R^3 \rightarrow R^2$ $T: R^3 \rightarrow R^2$ be a linear transformation, defined as $T(x, y, z) = (x + y, y + z)$

$T(x, y, z) = (x + y, y + z)$. Then which of the following options is /are true?

☐

$$T(0, 0, 0) = (0, 0)T(0, 0, 0) = (0, 0)$$

☐

$$T(1, 2, 3) = (3, 5)T(1, 2, 3) = (3, 5)$$

☐

$$T(0, 1, 1) = (1, 2)T(0, 1, 1) = (1, 2)$$

☐

$$T(1, 1, 1) = (3, 2)T(1, 1, 1) = (3, 2)$$

☐

T is injective.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$T(0, 0, 0) = (0, 0)T(0, 0, 0) = (0, 0)$$

$$T(1, 2, 3) = (3, 5)T(1, 2, 3) = (3, 5)$$

$$T(0, 1, 1) = (1, 2)T(0, 1, 1) = (1, 2)$$

1 point

Choose the correct set of options:

☐

T is a one to one linear transformation if and only if there does not exist any $v \neq 0 \in V$ $v \neq 0 \in V$ so that $T(v) = 0$ $T(v) = 0$.

☐

If $T: R^2 \rightarrow R^2$ $T: R^2 \rightarrow R^2$ is a surjective linear transformation, then it cannot be injective.

☐

If $T: R \rightarrow R$ $T: R \rightarrow R$ is a linear transformation which is not injective, then T must be 00, i.e., $T(v) = 0$ $T(v) = 0$ for all $v \in R$ $v \in R$.

☐

If there exists some non-zero vector $v \in V$ $v \in V$, such that $T(v) = 0$ $T(v) = 0$, for a linear transformation $T: V \rightarrow W$ $T: V \rightarrow W$, then T cannot be an isomorphism.

No, the answer is incorrect.

Score: 0

Accepted Answers:

T is a one to one linear transformation if and only if there does not exist any $v \neq 0 \in V$ $v \neq 0 \in V$ so that $T(v) = 0$ $T(v) = 0$.

If $T: R \rightarrow R$ $T: R \rightarrow R$ is a linear transformation which is not injective, then T must be 00, i.e., $T(v) = 0$ $T(v) = 0$ for all $v \in R$ $v \in R$.

If there exists some non-zero vector $v \in V$ $v \in V$, such that $T(v) = 0$ $T(v) = 0$, for a linear

transformation $T: V \rightarrow W$ $T: V \rightarrow W$, then T cannot be an isomorphism.

1 point

If $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is a linear transformation, such that $T(x, y) = (x, 0)$ $T(x, y) = (x, 0)$, then which of the following options is true?

☐

T is both one to one and onto.

☐

T is one to one, but not onto.

☐

T is onto, but not one to one.

☐

T is neither one to one nor onto.

No, the answer is incorrect.

Score: 0

Accepted Answers:

T is neither one to one nor onto.

1 point

If $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is a linear transformation, such that $T(x, y) = (x + y, x - y)$ $T(x, y) = (x + y, x - y)$, then which of the following options is true?

☐

T is both one to one and onto.

☐

T is one to one, but not onto.

☐

T is onto, but not one to one.

☐

T is neither one to one nor onto.

No, the answer is incorrect.

Score: 0

Accepted Answers:

T is both one to one and onto.

1 point

Let $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be a linear transformation, such that $T(2, 1) = (3, 2)$ $T(2, 1) = (3, 2)$ and $T(1, 3) = (0, 0)$ $T(1, 3) = (0, 0)$. Which of the following options is true?

☐

$T(x, y) = \frac{1}{5}(3x, 2y)$ $T(x, y) = 5(3x, 2y)$

☐

$T(x, y) = (x + 1, y + 1)$ $T(x, y) = (x + 1, y + 1)$

☐

$T(x, y) = \frac{1}{5}(9x - 3y, 6x - 2y)$ $T(x, y) = 5(9x - 3y, 6x - 2y)$

☐

T cannot be determined from the given information.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$T(x, y) = \frac{1}{5}(9x - 3y, 6x - 2y)$ $T(x, y) = 5(9x - 3y, 6x - 2y)$

1 point

Consider the linear transformation $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined as $T(v) = Av$ $T(v) = Av$, where $v = \begin{bmatrix} x \\ y \end{bmatrix}$, and $A = \begin{bmatrix} 3 & 2 \\ 4 & 5 \end{bmatrix}$ $v = [xy]$, and $A = \begin{bmatrix} 3 & 2 \\ 4 & 5 \end{bmatrix}$. Which of the following options is true?

☐

T is an isomorphism.

☐ T is one to one, but not onto.☐ T is onto, but not one to one.☐ T is neither one to one nor onto.**No, the answer is incorrect.****Score: 0****Accepted Answers:** T is an isomorphism.**Level 2***1 point*

Let $S: V_1 \rightarrow V_2$ and $T: V_2 \rightarrow V_3$ be two linear transformations. Let us define $T \circ S: V_1 \rightarrow V_3$ by $T \circ S(v) = T(S(v))$. Choose the correct set of options.

☐If $T \circ S$ is injective, then T must be injective.☐If $T \circ S$ is injective, then S must be injective.☐If $T \circ S$ is surjective, then T must be surjective.☐If $T \circ S$ is surjective, then S must be surjective.**No, the answer is incorrect.****Score: 0****Accepted Answers:**If $T \circ S$ is injective, then S must be injective.If $T \circ S$ is surjective, then T must be surjective.*1 point*If $T: R^2 \rightarrow R^3$ is a linear transformation defined by
$$T(x, y) = (2x + 3y, 5x - y, x + 6y)$$

Which of the following options is true?

☐ T is both one to one and onto.☐ T is one to one, but not onto.☐ T is onto, but not one to one.☐ T is neither one to one nor onto.**No, the answer is incorrect.****Score: 0****Accepted Answers:** T is one to one, but not onto.*1 point*

If $T: R^3 \rightarrow R^3$ is a linear transformation defined as follows, $T(x, y, z) = (a_1x + b_1y + c_1z, a_2x + b_2y + c_2z, a_3x + b_3y + c_3z)$. This can be represented as $T(v) = Av$, where

$$T(x, y, z) = (a_1x + b_1y + c_1z, a_2x + b_2y + c_2z, a_3x + b_3y + c_3z)$$

$A = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix}$, and $v = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$. Which of the following options

are true?

☐

TT is injective if and only if the system of linear equations $Av = 0$ has a unique solution.

☐

TT is injective if and only if $\text{rank}(A) = 3$.

☐

If the system of linear equations $Av = b$ for any $b \in \mathbb{R}^3$, has a solution, then TT must be surjective.

☐

If TT is not surjective, then there exists some $b \in \mathbb{R}^3$, such that the system of linear equations $Av = b$ has no solution.

No, the answer is incorrect.

Score: 0

Accepted Answers:

TT is injective if and only if the system of linear equations $Av = 0$ has a unique solution.

TT is injective if and only if $\text{rank}(A) = 3$.

If the system of linear equations $Av = b$ for any $b \in \mathbb{R}^3$, has a solution, then TT must be surjective.

If TT is not surjective, then there exists some $b \in \mathbb{R}^3$, such that the system of linear equations $Av = b$ has no solution.

[Check Answers](#)

Your score is: 0/12